Problem Set 2

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Due: October 22, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 22, 2026. No late assignments will be accepted.

Question 1: Political Science

The following table was created using the data from a study run in a major Latin American city.[1] As part of the experimental treatment in the study, one employee of the research team was chosen to make illegal left turns across traffic to draw the attention of the police officers on shift. Two employee drivers were upper class, two were lower class drivers, and the identity of the driver was randomly assigned per encounter. The researchers were interested in whether officers were more or less likely to solicit a bribe from drivers depending on their class (officers use phrases like, "We can solve this the easy way" to draw a bribe). The table below shows the resulting data.

[1] Fried, Lagunes, and Venkataramani (2010). "Corruption and Inequality at the Crossroad: A Multimethod Study of Bribery and Discrimination in Latin America. *Latin American Research Review.* 45 (1): 76-97.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	14	6	7
Lower class	7	7	1

(a) Calculate the χ^2 test statistic by hand/manually (even better if you can do "by hand" in R).

```r
# Question 1.a
# Creating table for observation for the data provided

bribe_table <- as.table(rbind(c(14,6,7), c(7,7,1)))
dimnames(bribe_table) <- list(
  Class = c("Upper Class", "Lower Class"),
  Bribe = c("Not Stopped", "Bribe Requested", "Stopped/Given Warning")
)

bribe_table

# Manually calculating the X^2

# Row sum for Upper and Lower Class
margin.table(bribe_table, 1)
# Total Upper Class = 27
# Total Lower Class = 15

# Column sum for Upper and Lower Class
margin.table(bribe_table,2)
# Not Stopped = 21
# Bribe Requested = 13
# Stopped/Given Warning = 8

```

```r
# Finding the F_Expected
expected <- outer(rowSums(bribe_table), colSums(bribe_table))/sum(bribe_table)
expected

# Chi Square Test
chi_sq <- sum((bribe_table - expected)^2 / expected)
chi_sq
# X Sqr = 3.791168

# Chi-Square Test in R
chisq.test((bribe_table))
# X Sqr is 3.7912
```

The Chi-Square by hand was determine to be 3.791. While the R function calculated the Chi-square to be the same as 3.791.

(b) Now calculate the p-value from the test statistic you just created (in R).[2] What do you conclude if $\alpha = 0.1$?

```r
# Question 1.b
# Calulating the P-Value

# 1) Degrees of Freedom Calculation
df <- (nrow(bribe_table)-1) * (ncol(bribe_table)-1)
df
# df = 2

# 2) P-Value

p_value <- pchisq(chi_sq,df,lower.tail = FALSE)
p_value
# P_Value = 0.1502306
```

We can conclude that with the p-value at 0.1502306, and alpha at 0.1, we can reject the null hypothesis.The value is right near .1, and we can also assume that the variables are dependent to each other.

[2]Remember frequency should be > 5 for all cells, but let's calculate the p-value here anyway.

(c) Calculate the standardized residuals for each cell and put them in the table below.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	0.3220306	-1.6419565	1.5230259
Lower class	-0.3220306	1.6419565	-1.5230259

```r
# Question 1.c
# Calculating the standardize residuals for each cell

sum_bt <- sum(bribe_table)
row_p <- rowSums(bribe_table)/sum_bt
col_p <- colSums(bribe_table)/sum_bt
exp <- outer(rowSums(bribe_table), colSums(bribe_table))/sum_bt

res <- (bribe_table -exp)/ sqrt(exp * outer(1 - row_p, 1-col_p))
res
```

(d) How might the standardized residuals help you interpret the results?

When observing the data from the residuals, you can notice that the lower class had more expediencies to have a bribe offer compare to those in the upper class. The use of this is to look for patterns.

Question 2: Economics

Chattopadhyay and Duflo were interested in whether women promote different policies than men.[3] Answering this question with observational data is pretty difficult due to potential confounding problems (e.g. the districts that choose female politicians are likely to systematically differ in other aspects too). Hence, they exploit a randomized policy experiment in India, where since the mid-1990s, $\frac{1}{3}$ of village council heads have been randomly reserved for women. A subset of the data from West Bengal can be found at the following link: `https://raw.githubusercontent.com/kosukeimai/qss/master/PREDICTION/women.csv`

Each observation in the data set represents a village and there are two villages associated with one GP (i.e. a level of government is called "GP"). Figure 1 below shows the names and descriptions of the variables in the dataset. The authors hypothesize that female politicians are more likely to support policies female voters want. Researchers found that more women complain about the quality of drinking water than men. You need to estimate the effect of the reservation policy on the number of new or repaired drinking water facilities in the villages.

Figure 1: Names and description of variables from Chattopadhyay and Duflo (2004).

[3]Chattopadhyay and Duflo. (2004). "Women as Policy Makers: Evidence from a Randomized Policy Experiment in India. *Econometrica*. 72 (5), 1409-1443.

(a) State a null and alternative (two-tailed) hypothesis.

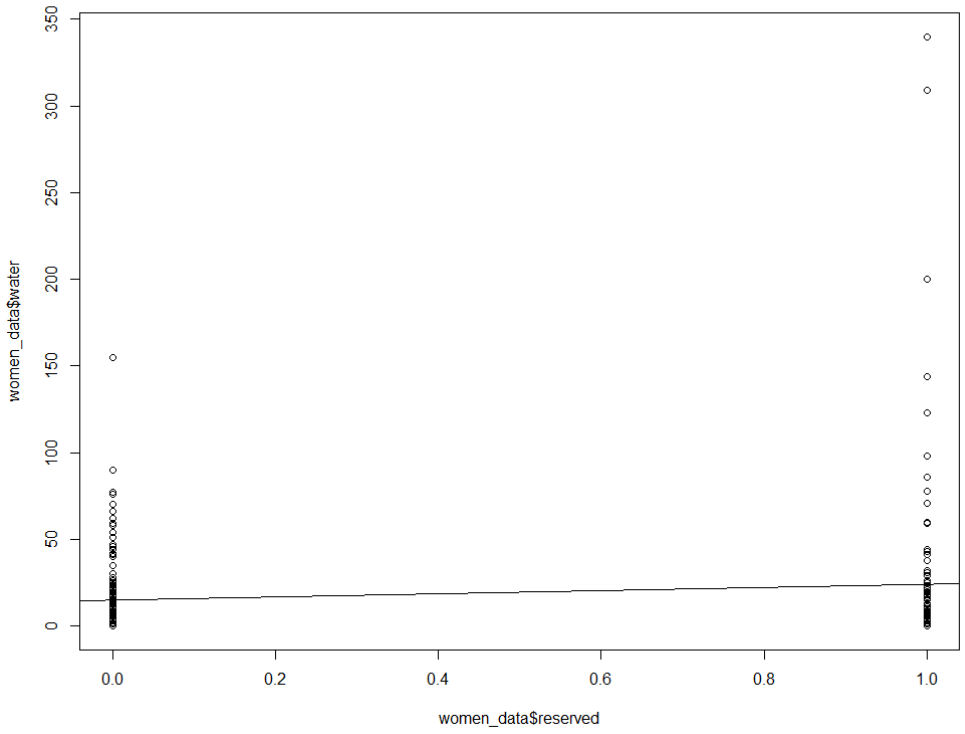
H0 is that female politicians are not likely to support policies female voter want than more than men. (I.e, gender does not determine the out come for the vote issued by the female politician) HA is that females politicians are likely to support female voter wants.

(b) Run a bivariate regression to test this hypothesis in R (include your code!).

```
# Data
women_data <- read.csv("https://raw.githubusercontent.com/kosukeimai/qss/
    master/PREDICTION/women.csv")
women_data
```

```
model <-lm(water~reserved, data = women_data)
summary(model)

# Plotting
plot(women_data$reserved,women_data$water)
abline(model)
```



(c) Interpret the coefficient estimate for reservation policy.

The model that I created use the variables water and reserved. When a village had a reserved leadership for women, the increase of the water increased by 9.252423. Therefore, we can suggest that when there are women leaders in, office the increase of women's demands are reflected. We can also look at the p value of 0.0197, we can reject the null hypothesis that suggest that female leaders may not take gender into considerations.